



PRAYCG Protocol Module Updates

PRAYCG4, structural controls, confound defenses, Williams sequences, and Semantic Meaning Gradient

Purpose

A standalone guide to the protocol architecture added after the original PRAYCG3 runner. It explains what each addition changes, which alternative explanation it addresses, and what it does not establish.

Component	Current role
PRAYCG3 v3.0	Preserved three-prong reference protocol; fixed public order remains available.
PRAYCG4 v4.0	Adds ShotOrderScramble between PhaseScrambled and Target.
Prospective sequence package v0.1	Adds Williams-sequence variants for period and first-order carryover balance.
Semantic Meaning Gradient v1.0	Adds a separate conditional protocol for coherent semantic-zero, high-meaning Target, and arithmetic Override.
Confound-defense family	Adds HOC-R, OSA, OHC, AAM, RespDualPath, DGA/context, and stronger provenance/timing checks.

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Document status: public-facing protocol specification. Protocol availability does not itself establish construct validity.

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1. What changed and why

ORIENTATION

The protocol update is not one new score. It is a change in experimental architecture. PRAYCG now separates three questions that the earlier three-prong design could not cleanly separate:

Question	Protocol comparison	What it constrains
Is the effect larger for intact material than a strongly structure-damaged control?	Target - PhaseScrambled	Low-level sensory entrainment, but with substantial structural mismatch remaining.
Is the effect related to temporal/narrative order beyond locally intact shots?	Target - ShotOrderScramble	Cross-shot sequence while retaining more faces, bodies, objects, voices, and biological motion.
Is the effect changed by analytic extraction during identical intact media?	Target - Contextual Override	Task stance, cue monitoring, and extraction load.

The protocol library then makes those designs selectable without rewriting the shared PsychoPy execution engine. Counterbalanced variants address systematic order imbalance, and the confound modules challenge specific alternatives before an effect is described as narrative reception or persistent state change.

Core logic

No single control is expected to solve every attribution problem. Confidence comes from convergent contrasts with different failure modes, plus explicit quality and confound gates.

1.1 Protocol update map

Layer	Addition	Reason
Acquisition	Plug-and-play protocol library	Select PRAYCG3, PRAYCG4, or SMG while preserving one runner framework.
Structural control	ShotOrderScramble	Retain local high-order content while disrupting larger temporal organization.
Order control	Williams variants	Reduce systematic period and immediate carryover imbalance.
Semantic manipulation	SMG protocol	Test a declared meaning gradient without treating randomness as semantic zero.
Attribution	HOC-R/OSA/OHC/AAM/RespDualPath/DGA	Challenge stimulus, gaze, order, afterglow, respiration, access, and task explanations.
Integrity	Hashes, manifests, ALS script-location test	Verify what ran, what was shown, and whether timing evidence belongs to the expected executable path.

2. PRAYCG4: why the fourth branch is necessary

NEW STRUCTURAL-CONTROL PROTOCOL

2.1 The limitation of PhaseScrambling alone

PhaseScrambling is useful because it damages recognizable narrative organization while preserving portions of the amplitude spectrum and gross audiovisual energy. But it also changes or destroys many features that are not merely narrative order: recognizable faces, contours, objects, bodies, speech structure, natural motion, and scene coherence.

Therefore, a Target - Phase difference can reflect intact meaning, but it can also reflect the much broader fact that one stimulus looks and sounds like natural media while the other does not. This does not make the Phase control invalid. It makes the contrast broad.

```
Target - PhaseScrambled
= narrative/temporal organization
+ local object/face/body structure
+ natural motion and speech structure
+ familiarity and recognizability
+ any remaining unmatched sensory or artifact effects
```

2.2 What ShotOrder changes

ShotOrderScramble begins from the intact Target, divides it into declared shots, and permutes those shots. Within each retained shot, local pixels, faces, bodies, objects, voices, and biological motion remain as intact as the editing process permits. What changes is their cross-shot temporal order.

```
V_Target = S_1 (+) S_2 (+) ... (+) S_m
V_ShotOrder = S_pi(1) (+) S_pi(2) (+) ... (+) S_pi(m)
```

Here pi is a recorded permutation of the m shot indices and (+) denotes ordered concatenation. The manipulation is designed to reduce larger-scale sequence coherence while preserving more local high-order visual and auditory content than PhaseScrambling.

2.3 What the new contrast asks

```
Delta_TS = mean(Y_Target) - mean(Y_ShotOrder)
```

A positive Target - ShotOrder contrast is evidence that the measured feature differs when the same local shot material is presented in coherent versus disrupted order. It is not, by itself, proof of meaning. New cuts, audio discontinuities, predictability, familiarity, attention, and editing artifacts remain plausible contributors.

Why PRAYCG4 is necessary

It creates a narrower structural question between the very broad Phase control and the task-based Override control. Without it, local high-order content and larger narrative-temporal organization are more difficult to distinguish.

3. PRAYCG4 protocol specification

IMPLEMENTED FIXED-ORDER PROTOCOL

3.1 Public fixed-order sequence

```
BASELINE_1
-> PHASE_SCRAMBLED -> WASHOUT/report
-> SHOT_ORDER_SCRAMBLE -> WASHOUT/report
-> TARGET -> WASHOUT/report
-> CONTEXTUAL_OVERRIDE -> WASHOUT/report
-> BASELINE_2_REFLECTION -> FINAL_REPORT
```

PRAYCG4 preserves Baseline 1, a washout and branch report after every stimulus condition, Baseline 2/final reflection, and the final report. ShotOrder is inserted only after the completed PhaseScrambled washout/report. It does not replace the Phase control.

3.2 Primary structural contrasts

```
Delta_TP = mean(Y_Target) - mean(Y_PhaseScrambled)
Delta_TSO = mean(Y_Target) - mean(Y_ShotOrderScramble)
Delta_T0 = mean(Y_Target) - mean(Y_ContextualOverride)
```

Contrast	Interpretive target	Still not ruled out
Target - Phase	Intact structure versus strongly damaged structure	Broad naturalness, recognizability, local content, motion, and speech differences.
Target - ShotOrder	Coherent cross-shot order beyond locally intact shots	Cut rhythm, discontinuity, order expectancy, familiarity, and attention.
Target - Override	Natural viewing versus analytic extraction during identical intact media	Cue burden, arithmetic demand, gaze, compliance, and task relief.

3.3 ShotOrder provenance

- Record the clean master path and SHA-256 hash.
- Record every shot boundary and the rule used to define it.
- Record the permutation seed, explicit permutation, editor settings, and audio treatment.
- Hash the generated ShotOrder derivative and its manifest.
- Verify duration, cue schedule, ALS pulse placement, frame rate, audio continuity, and output readability.
- Preserve the exact derivative selected by the runner in the run manifest.

```
h_f = SHA256(bytes(f))
```

Identity is not validity
A hash proves that the same bytes were used. It does not prove that the manipulation isolated the intended scientific construct.

3.4 Branch reports and independent outcomes

Each branch report should preserve meaning, absorption, emotional afterglow, story-active carryover, task/extraction burden, and confound reporting. Recognition, comprehension, confidence, and narrative-coherence outcomes should be kept independent of the analysis index they are intended to validate.

4. The new confound-defense modules

ATTRIBUTION SAFEGUARDS

The confound modules do not manufacture a positive interpretation. Their job is to make a positive interpretation harder to earn and easier to qualify. Each module addresses a different alternative explanation.

4.1 HOC-R: high-order confound residualization

HOC-R asks whether a physiological or analysis feature can be predicted from measurable properties of the media rather than from the proposed narrative construct. The design matrix may include low-level variables such as luminance, audio envelope, cuts, optical flow, and cue timing, plus high-order variables such as face presence, body presence, object density, motion category, speech, and area-of-interest occupancy.

```
y = X beta + epsilon
y_hat = X beta_hat
y_resid = y - y_hat
R2_train = 1 - sum(y-y_hat)^2 / sum(y-mean(y))^2
```

For interpretation, held-out or cross-validated prediction is more important than training fit. A large training R-squared can reflect overfitting. HOC-R therefore reports regressor availability, collinearity, training fit, held-out behavior where supportable, and whether residualization is scientifically defensible.

HOC-R result	Meaning
Required regressors present; held-out prediction stable	Residualization may be informative, with uncertainty and model assumptions reported.
Only low-level regressors present	Target - Phase remains cautioned because faces, bodies, objects, and other high-order structure are unmodeled.
Regressors missing or collinear	Do not treat residual output as a clean narrative signal.
Stimulus model explains most of the candidate feature	The result is more parsimoniously described as stimulus tracking unless independent evidence survives.

HOC-R boundary
Residualization can constrain a measured alternative; it cannot prove that the remaining variance is meaning. Unmeasured confounds remain unmeasured.

4.2 OSA: ocular and spatial-attention burden

OSA addresses the possibility that cue placement, small text, eye strain, squinting, gaze revisits, or spatial competition changes EEG and behavior independently of narrative reception.

```
cue_duty = display_duration / cue_interval
report_burden = mean_available(legibility, blur, eye_strain, stall, task_load)
OSA = mean_available(cue_duty, report_burden)
```

The current caution threshold of OSA $\geq .35$ when gaze or area-of-interest data are absent is exploratory. It is a warning rule, not a probability of ocular confounding.

4.3 OHC: order, habituation, and carryover

OHC records the branch order and challenges the assumption that a brief washout produces a full biological reset. It treats period, fatigue, habituation, sensitization, and prior-condition carryover as design risks. Its scores are heuristic warnings, not estimated probabilities.

```
Observed contrast(a,b)
= condition(a-b) + period(a-b)
+ carryover_into_a - carryover_into_b + error
```

The prospective Williams variants reduce systematic first-order imbalance, but they do not eliminate long-memory carryover, fatigue, or nonlinear order effects.

4.4 AAM: afterglow attribution model

AAM asks whether Baseline 2 and late reports look more like continuing Target afterglow or relief/recovery from the Override task. This matters because Baseline 2 occurs after all prior conditions and is therefore cumulative rather than Target-only.

```
sim(x,y) = dot(x,y) / (||x|| ||y||)
NAI = mean_available(sim(B2,W2), target_echo, target_afterglow, story_active_W2)
TCRI = mean_available(task_load, compliance, stall, sim(B2,W3))
AAM = NAI - TCRI
```

Under the current heuristic, AAM $> .20$ is Target-afterglow-like, AAM $< -.20$ is task-relief-like, and the middle is mixed. These cutoffs are exploratory.

4.5 RespDualPath and continuous autonomic checks

RespDualPath tests whether apparent state changes coincide with changes in breathing, motion, cardiac timing, or respiratory-cardiac coupling. The legacy path preserves abrupt respiratory change events from feature tables. The continuous path derives beat events, HRV, respiration phase/depth/rate, sigh and hold candidates, motion cautions, and time-shift nulls from raw streams.

```
d_t = |resp_t - resp_(t-1)|
legacy threshold = mean(d) + 2.5 SD(d)
r(lag;t) = corr_wt(HR(u), respiration(u+lag))
p_shift = [1 + count(|r_null| >= |r_observed|)] / (B+1)
```

Autonomic boundary

Autonomic timing can support or challenge an interpretation, but it remains separate from primary CAI. HRV or respiratory change does not prove meaning, regulation, or parasympathetic truth.

4.6 DGA and contextual gating

DGA separates semantic access from emotional/integrative impact and task burden. A Target can be absorbing despite degraded audio, while an Override can improve semantic access yet increase analytic extraction load. DGA explains branch availability; it does not replace the primary endpoint.

$$D_{gate,b} = \text{sigmoid}(2.0 \text{ SA}_b + 1.4 \text{ DA}_b + 1.1 \text{ EI}_b - 1.2 \text{ X}_b - 1.8 \text{ C}_b - 0.75)$$

Term	Role
SA	Semantic access after comprehension, volume, noise, and sync difficulty.
DA	Decoder availability from meaning, absorption, afterglow, and story-active reports.
EI	Emotional/integrative impact.
X	Extraction or task load.
C	Confound burden.

When used with CAI, contextual reliability remains a separate sensitivity output:

$$\begin{aligned} \text{Reliability_context} &= D_{gate} * (1 - \text{confound_load}) * (1 - \text{extraction_load}) \\ \text{CAI_context_gated} &= \text{CAI_core} * \text{Reliability_context} \end{aligned}$$

4.7 CET-R and the sensory-environment model

CET-R models how much a candidate physiological feature tracks exogenous stimulus variables. It is especially important for Phase and ShotOrder comparisons because both manipulations change temporal stimulus structure.

$$\begin{aligned} Y_{feature}(t) &= \beta_0 + \beta_u u(t-lag) + \beta_{art} \text{Artifact}(t) + \epsilon(t) \\ Y_{resid}(t) &= \epsilon(t) \end{aligned}$$

A residual is only as clean as the regressors and validation. If the stimulus model is incomplete, the residual can still contain unmeasured stimulus effects.

4.8 How the defenses work together

Alternative explanation	Primary defense
Low-level audiovisual drive	Phase control, StimulusFingerprint, CET-R
Local faces/bodies/objects and natural motion	ShotOrder control, HOC-R high-order regressors
Cue location, squint, and gaze burden	OSA, CVB, SquintProxy, gaze/AOI where available
Period, habituation, and carryover	OHC and Williams sequence variants
Baseline 2 as task relief rather than Target echo	AAM and branch-specific reports
Respiration/cardiac explanation	RespDualPath and continuous autonomic module
Poor comprehension or decoder availability	DGA and independent comprehension/confidence outcomes

5. Williams sequences and prospective counterbalancing

PROSPECTIVE DESIGN PACKAGE

5.1 Why the fixed public order is not enough for confirmation

The fixed PRAYCG3 and PRAYCG4 runners are valuable for continuity and engineering, but condition and position are linked. In PRAYCG4, for example, Phase always occurs before ShotOrder, ShotOrder before Target, and Target before Override. A difference between branches therefore includes any systematic difference caused by period, fatigue, adaptation, expectation, or carryover.

$$\begin{aligned} \Delta_{\hat{a,b}} = & [\tau_a - \tau_b] + [\text{period}_a - \text{period}_b] \\ & + [\text{carryover_into}_a - \text{carryover_into}_b] + \text{error} \end{aligned}$$

5.2 Williams-sequence objective

A Williams design rotates conditions so that each condition appears approximately equally often in each period and each ordered pair appears approximately equally often as immediate neighbors. For an even number q of conditions, q sequences are sufficient. For an odd number, the reversed set is also used, producing $2q$ sequences.

Period balance: $N(\text{period}, \text{condition})$ approximately constant
First-order carryover balance: $N(a \rightarrow b)$ approximately constant for $a \neq b$

What Williams balancing does not do
It does not erase long-lasting biological carryover, fatigue, learning, nonlinear sequence effects, or stimulus-specific interactions. It makes first-order imbalance less systematic and more modelable.

5.3 PRAYCG4 prospective variants

Variant	Condition order
4P-01	Phase -> ShotOrder -> Override -> Target
4P-02	ShotOrder -> Target -> Phase -> Override
4P-03	Target -> Override -> ShotOrder -> Phase
4P-04	Override -> Phase -> Target -> ShotOrder

Baseline 1 remains before the first condition. Every condition retains its washout/report. Baseline 2/final reflection and the final report remain after the final assigned condition.

5.4 PRAYCG3 prospective variants

PRAYCG3 has three conditions, so the forward Williams set is paired with its reverse. The public fixed-order runner is still Phase -> Target -> Override; the following are separately named prospective variants.

Variant	Condition order
3P-01	Phase -> Target -> Override
3P-02	Target -> Override -> Phase
3P-03	Override -> Phase -> Target
3P-04	Override -> Target -> Phase
3P-05	Phase -> Override -> Target
3P-06	Target -> Phase -> Override

5.5 Deterministic assignment

$$j(p,s) = ((p-1) + (s-1)) \bmod m$$

The template can assign participant p and stimulus s to one of m sequence variants without looking at outcome data. The assignment rule, participant identifier policy, stimulus identifier, and selected variant must be frozen before acquisition.

- Do not manually choose a favorable sequence after seeing participant or stimulus responses.
- Record the assigned variant in the runner manifest and hash the locked configuration.
- Model period and, when supportable, carryover terms even after counterbalancing.
- Report incomplete cells, dropouts, reruns, and deviations from the assignment rule.
- Use blinded feasibility data for burden and variance planning, not for choosing the condition that wins.

5.6 Practical deployment

Stage	Required action
Before study	Freeze the eligible sequence set and deterministic assignment rule.
Before participant	Resolve participant/stimulus IDs, select variant, lock runner, record hashes.
During run	Preserve branch markers, washouts/reports, ALS events, and deviation log.
Analysis	Use canonical condition labels plus period, sequence, participant, and stimulus identifiers.
Reporting	Show cell counts and any imbalance; do not imply counterbalancing eliminated carryover.

6. Semantic Meaning Gradient protocol

CONDITIONAL PROTOCOL MODULE

6.1 Scientific purpose

SMG asks whether a declared gradient of semantic and human relevance can be manipulated within a protocol. It is separate from PRAYCG3 and PRAYCG4 because it tests a different design question. PRAYCG4 asks about intact temporal order versus structural controls; SMG asks whether a coherent low-meaning condition, a high-meaning Target, and an analytic Override differ in the expected direction.

SEMANTIC_ZERO_1 -> HIGH_MEANING_TARGET_1 -> ARITHMETIC_OVERRIDE_1

6.2 Why Semantic-Zero is not PhaseScrambled

A random or PhaseScrambled stimulus is not automatically semantically zero. It can create novelty, confusion, threat, prediction error, visual search, or puzzle demand. SMG therefore requires a distinct coherent stimulus designed to be low on the declared meaning dimensions while remaining watchable and structurally intelligible.

Manipulation rule
Semantic-Zero is an independently selected coherent low-meaning condition. It cannot be silently replaced with the PRAYCG Phase control.

6.3 Proposed semantic-density function

$$\begin{aligned} \text{SMD}(t) = & \text{sigmoid}[w_H H(t) + w_A A(t) + w_V V(t) + w_C C(t) \\ & + w_R R(t) + w_K K(t) + w_T T(t) \\ & - w_P \text{Puzzle}(t) - w_O \text{Overload}(t)] \end{aligned}$$

Symbol	Proposed dimension
H	Human or social presence
A	Agency
V	Vulnerability
C	Causal continuity
R	Repair or care
K	Stakes
T	Transformation
Puzzle	Puzzle or extraction demand
Overload	Overload unrelated to intended meaning

Weights must be nonnegative, normalized, and frozen independently of the physiology used to test the theory. Otherwise the semantic score can be tuned to reproduce the neural result it is supposed to predict.

6.4 Summary and temporal-arc quantities

$$M_{total} = (1/T) \int_0^T SMD(t) dt$$

$$M_{peak} = \max_t SMD(t)$$

$$M_{arc_star} = a \text{Var}[SMD(t)] + b T_0 \max_t |dSMD(t)/dt|$$

Variance and rate of change have different units. The time scale T_0 is therefore required before combining them. Without an explicit time scale, variance and maximum slope should remain separate reported quantities.

6.5 Manipulation-check candidates

```

SZI = (1-Meaning)(1-Empathy)(1-Absorption)(1-Threat)(1-PuzzleDemand)
OPI = TaskCompliance * AnalyticEffort * (1-Empathy)
      * (1-Meaning) * (1-NarrativeEngagement)
DeltaM = M_target - M_semantic_zero
MCS = mean(StoryActive, EmotionalAfterglow, Replay, FeltReturnDelay)

```

Quantity	Intended use	Current boundary
SZI	Whether Semantic-Zero was low on declared subjective dimensions	Proposed manipulation check; mapping and cutoffs unvalidated.
OPI	Whether arithmetic Override produced compliant analytic extraction	Proposed manipulation check; task success and burden must be independent.
DeltaM	Target minus Semantic-Zero meaning separation	Depends on declared and validated meaning measurement.
MCS	Carryover after the high-meaning condition	Cannot assign cumulative late state uniquely to Target without order controls.
MRI	Proposed composite in the theory document	Recorded concept only; current runner does not compute it.

All report inputs must be explicitly mapped to [0,1]. Existing suggested thresholds are development candidates rather than validated decision boundaries.

6.6 Required report domains

- Meaning, empathy, absorption, threat, puzzle demand, and narrative engagement.
- Carryover, replay, emotional afterglow, story-active state, and felt return delay.
- Task extraction, analytic effort, arithmetic compliance, and confidence.
- Independent recognition and comprehension rather than embedding those outcomes in the tested index.
- Confounds including familiarity, sensory problems, fatigue, prior exposure, and current-life relevance.

SMG status

SMG v1.0 is a conditional acquisition module and theory-testing scaffold. Its semantic indices and thresholds remain proposed and unvalidated.

7. Protocol modules in the Control Center

OPERATIONAL WORKFLOW

7.1 Plug-and-play architecture

The Control Center separates the shared runner engine from the selected protocol module. A protocol definition supplies the condition order, required media inputs, marker vocabulary, branch reports, washouts, baselines, final report, and readiness rules. The shared launcher then starts PsychoPy with the selected locked protocol.

```
Protocol selection
-> input binding
-> readiness check
-> configuration lock and hashes
-> PsychoPy + selected runner
-> markers, reports, and manifest
```

7.2 Recommended operator workflow

Step	Action	Evidence produced
1	Choose PRAYCG3, PRAYCG4, or SMG from the protocol library.	Protocol ID and version.
2	Select the master stimulus and required derivatives/anchors.	Resolved paths and hashes.
3	Confirm required conditions and reports are present.	Readiness PASS/CAUTION/INELIGIBLE reasons.
4	Run signal-quality/contact and optional 30-second preflight.	Contact score and preflight validation report.
5	Run the ALS barcode placement script-location test.	Expected script, launch path, visible pulse, recovered timing relationship.
6	Lock the runner/configuration and launch PsychoPy.	Frozen configuration and code/config hashes.
7	Complete every washout/report, Baseline 2, and final report.	Independent report and confound records.
8	Analyze using canonical condition and block-instance identifiers.	Status-aware outputs with warnings and evidence grades.

7.3 Readiness requirements by protocol

Protocol	Distinct required inputs
PRAYCG3	PhaseScrambled, Target/Override identity, Baseline 1/2, reports, ALS/timing, provenance.
PRAYCG4	All PRAYCG3 requirements plus ShotOrder derivative, shot manifest, permutation, and hash.
SMG	Distinct coherent Semantic-Zero, high-meaning Target, arithmetic Override, ratings, and declared semantic metadata.

Locking rule

The runner should record the chosen protocol module before acquisition. Changing a condition, stimulus, order, or scoring configuration after outcome inspection creates a new version rather than silently updating the locked run.

8. What changed in PRAYCG3

LIMITED REFERENCE NOTE

The scientific order of the public PRAYCG3 runner was not redesigned. Its role is to preserve the original three-prong architecture and comparability with prior runs.

```
BASELINE_1
-> PHASE_SCRAMBLER -> washout/report
-> TARGET -> washout/report
-> CONTEXTUAL_OVERRIDE -> washout/report
-> BASELINE_2_REFLECTION -> final report
```

8.1 Changes around PRAYCG3 rather than inside it

- PRAYCG3 is now a selectable protocol module in the shared library.
- Baseline 1, every washout/report, Baseline 2/final reflection, and the final report remain explicit required stages.
- Media, anchors, runner configuration, and derivatives receive path/hash provenance.
- ALS barcode placement and script-location verification is available as an acquisition-integrity test.
- The six prospective Williams variants are separately named; they do not silently change the public fixed-order runner.
- The expanded confound-defense family can audit PRAYCG3 data where the necessary inputs exist.

PRAYCG3 continuity

The update formalizes packaging, provenance, readiness, and prospective order variants. It does not remove Baseline 1, Baseline 2, washouts, branch reports, or the final report.

9. Prospective use, interpretation, and claim boundaries

STUDY GOVERNANCE

9.1 Freeze before confirmatory review

- Select the protocol and sequence-assignment rule.
- Freeze stimuli, ShotOrder manifests, anchors, reports, timing rules, and hashes.
- Declare primary contrasts and whether the endpoint is A-MRED, DeltaMRED, or another preregistered measure.
- Freeze exclusions, coverage thresholds, lag windows, nulls, multiplicity correction, and random effects.
- Declare how period and carryover will be modeled.
- Declare independent recognition, comprehension, confidence, and narrative-coherence outcomes.
- Register the protocol and preserve the exact code/configuration hashes before inspecting confirmatory outcomes.

9.2 Recommended mixed model

```
Y_(participant,stimulus,condition)
= beta_0 + beta_condition + beta_period
+ u_participant + v_stimulus + error
u_participant ~ N(0,sigma_u^2)
v_stimulus ~ N(0,sigma_v^2)
```

A carryover term may be added if the design, sample size, and pre-registration support it. Counterbalancing is not a substitute for modeling and transparent cell counts.

9.3 Public-language boundary

Permitted	Not permitted
Target differed from ShotOrder under the frozen model.	ShotOrder proved narrative meaning.
The Phase contrast survived declared sensory regressors.	PhaseScrambling removed every non-meaning difference.
HOC-R reduced or did not reduce the candidate effect.	Residual variance must be meaning.
Williams variants reduced systematic first-order imbalance.	Counterbalancing eliminated carryover.
SMG manipulation checks separated the declared conditions.	SMG is a validated universal meaning scale.
Respiration did or did not explain the measured timing pattern.	Autonomic change proves narrative reception.

Overall interpretation
PRAYCG4 and the confound modules strengthen attribution by narrowing alternatives. They do not convert a single-subject or exploratory result into proof. The value is a more falsifiable chain of questions.

10. Protocol selection and analysis quick reference

OPERATOR SUMMARY

If the research question is...	Use...	Primary comparison
Does intact natural viewing differ from a strong structure-damaged control and an analytic task?	PRAYCG3	Target-Phase and Target-Override
Does coherent temporal order matter beyond locally intact shots?	PRAYCG4	Target-Phase, Target-ShotOrder, Target-Override
Does a declared coherent semantic gradient behave as predicted?	SMG	HighMeaning-SemanticZero and HighMeaning-ArithmeticOverride
Is period/carryover balance required for prospective inference?	Prospective Williams variant	Same protocol contrasts with sequence and period recorded

10.1 Minimum PRAYCG4 audit package

- Locked PRAYCG4 runner/configuration and code hash.
- Target, PhaseScrambled, ShotOrder, and Override identity/provenance records.
- Shot boundary/permutation manifest and derivative hash.
- Baseline 1, all branch markers, every washout/report, Baseline 2, and final report.
- ALS timing status and script-location test result.
- Artifact, CET-R, HOC-R availability, OSA, OHC, AAM, and RespDualPath status.
- Canonical condition, block-instance, sequence, period, participant, and stimulus identifiers.
- Declared primary contrasts, exclusions, missingness, multiplicity, and evidence grade.

10.2 Selected methodological references

- Williams, E. J. (1949). Experimental Designs Balanced for the Estimation of Residual Effects of Treatments. Australian Journal of Scientific Research, Series A 2:149-168. DOI 10.1071/CH9490149.
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Final summary

PRAYCG4 adds the missing middle control between PhaseScrambling and intact Target viewing. Williams variants improve prospective order balance. SMG opens a separate semantic-gradient protocol. HOC-R and the other confound modules make every positive result answer a harder set of alternative explanations before stronger language is permitted.